

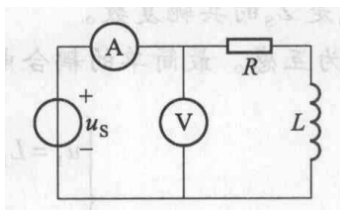
答案 4.2

4.2 写出下列电压、电流相量所代表的正弦电压和电流(设角频率为 ω): (a) $\dot{U}_m = 10 \angle -10^\circ \text{ V}$

$$u = 10 \cos(\omega t - 10^\circ) \text{ V}$$

答案 4.3

4.3 图示电路中正弦电流的频率为 50 Hz 时,电压表和电流表的读数分别为 100 V 和 15 A;当频率为 100 Hz 时,读数为 100 V 和 10 A。试求电阻 R 和电感 L 。



解: 电压表和电流表读数为有效值, 其比值为阻抗模, 即

$$\sqrt{R^2 + (\omega L)^2} = U / I$$

将已知条件代入, 得

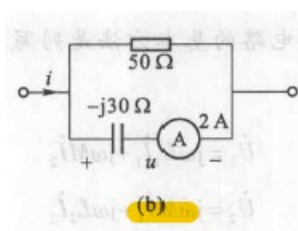
$$\begin{cases} \sqrt{R^2 + (2\pi \times 50 \times L)^2} = \frac{100\text{V}}{15\text{A}} \\ \sqrt{R^2 + (2\pi \times 100 \times L)^2} = \frac{100\text{V}}{10\text{A}} \end{cases}$$

联立方程, 解得

$$L = 13.7\text{mH}, R = 5.08\Omega$$

答案 4.4

4.4 图示各电路中已标明电压表和电流表的读数, 试求电压 u 和电流 i 的有效值。



$$u = -j30\Omega * 2\text{A} = -j60 \text{ V}$$

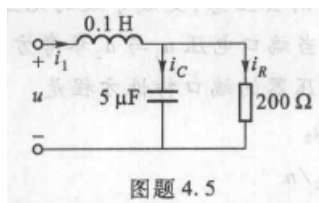
$$U = 60\text{V}$$

$$i = \frac{u}{50\Omega} + 2\text{A} = \left(2 - j\frac{6}{5}\right) \text{ A}$$

$$I = \sqrt{2^2 + 1.2^2} = 2.33 \text{ A}$$

答案 4.5

4.5 在图示电路中已知 $i_R = \sqrt{2} \cos \omega t$ A, $\omega = 2 \times 10^3$ rad/s。求各元件的电压、电流及电源电压 u , 并作各电压、电流的相量图。



图题 4.5

$$u_R = u_C = i_R * 200 \Omega = 200\sqrt{2} \cos \omega t \text{ V}$$

$$i_C = \frac{u_C}{\frac{1}{j\omega C}} = 2\sqrt{2} \cos(\omega t + 90^\circ) \text{ A}$$

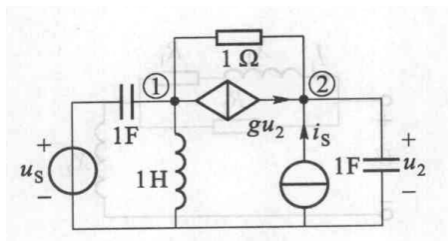
$$i_1 = i_C + i_R = \sqrt{10} \cos(\omega t + 63.43^\circ) \text{ A}$$

$$u_L = i_1 * j\omega L = 200\sqrt{10} \cos(\omega t + 153.43^\circ) \text{ V}$$

$$u = u_L + u_C = 400 \cos(\omega t + 135^\circ) \text{ V}$$

答案 4.7

4.7 已知图示电路中 $g = 1$ S, $u_s = 10\sqrt{2} \cos \omega t$ V, $i_s = 10\sqrt{2} \cos \omega t$ A, $\omega = 1$ rad/s。求受控电流源的电压 u_{12} 。



解：电压源和电流源的相量分别为

$$\dot{U}_s = 10 \angle 0^\circ \text{ V}, \quad \dot{I}_s = 10 \angle 0^\circ \text{ A}$$

对节点①和②列相量形式节点电压方程

$$\begin{cases} (j\omega C_1 + \frac{1}{j\omega L} + 1S)\dot{U}_{n1} - 1S \times \dot{U}_{n2} = j\omega C_1 \dot{U}_s - g\dot{U}_2 \\ -1S \times \dot{U}_{n1} + (j\omega C_2 + 1S)\dot{U}_{n2} = \dot{I}_s + g\dot{U}_2 \end{cases}$$

由图可知受控源控制量

$$\dot{U}_2 = \dot{U}_{n2}$$

解得

$$\dot{U}_{n1} = j10 \text{ V} \quad \dot{U}_{n2} = 10 - j10 \text{ V}$$

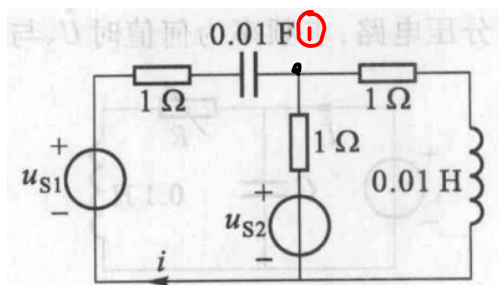
$$\dot{U}_{12} = \dot{U}_{n1} - \dot{U}_{n2} = (-10 + j20) \text{ V} = 22.36 \angle 116.57^\circ \text{ V}$$

受控电流源的电压为

$$u_{12} = 22.36\sqrt{2} \cos(\omega t + 116.57^\circ) \text{ V}$$

答案 4.10

4.10 已知图示电路中 $u_{s1} = u_{s2} = 4 \cos \omega t \text{ V}$, $\omega = 100 \text{ rad/s}$ 。试求电流 i 。



解：图示电路容抗

$$X_C = -\frac{1}{\omega C} = -\frac{1}{100 \times 0.01} \Omega = -1 \Omega,$$

感抗

$$X_L = \omega L = (100 \times 0.01) \Omega = 1 \Omega$$

列节点电压方程

$$\left[\frac{1}{1\Omega + j(-1\Omega)} + \frac{1}{1\Omega} + \frac{1}{1\Omega + j\Omega} \right] \dot{U}_{n1} = \frac{\dot{U}_{s1}}{1\Omega + j(-1\Omega)} + \frac{\dot{U}_{s2}}{1\Omega} \quad (1)$$

将

$$\dot{U}_{s1} = \dot{U}_{s2} = 2\sqrt{2} \angle 0^\circ \text{ V 代入(1)式}$$

解得

$$\dot{U}_{n1} = \sqrt{5} \angle 18.43^\circ \text{ V}$$

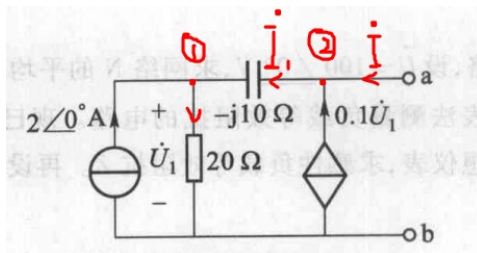
$$\dot{i} = -\frac{-\dot{U}_{n1} + \dot{U}_{s1}}{1\Omega + j(-1\Omega)} = \frac{\sqrt{2}}{2} \text{ A}$$

电流

$$i = \cos(100t) \text{ A}$$

答案 4.12

4.12 求图示电路的戴维宁等效电路。



解: (1)求开路电压 $\dot{U}_{oc}$

对图(a)电路列节点电压方程

$$\begin{cases} (\frac{1}{20} + \frac{1}{-j10})S \times \dot{U}_{n1} - \frac{1}{-j10} \times \dot{U}_{n2} = 2\angle 0^\circ A & (1) \\ -\frac{1}{-j10} S \times \dot{U}_{n1} + \frac{1}{-j10} S \times \dot{U}_{n2} = 0.1S \times \dot{U}_1 & (2) \end{cases}$$

受控源控制量 $\dot{U}_1$ 即为节点电压 $\dot{U}_{n1}$, 即

$$\dot{U}_1 = \dot{U}_{n1} \quad (3)$$

将式(3)代入式(2)再与式(1)联立解得

$$\dot{U}_{n1} = -40V, \quad \dot{U}_{n2} = \dot{U}_{oc} = 40\sqrt{2}\angle 135^\circ V$$

(2)求等效阻抗 Z_i

在 ab 端外施电压源 $\dot{U}_{ab}$, 求输入电流 $\dot{I}$, $\dot{U}_{ab}$ 与 $\dot{I}$ 的比值即为等效阻抗 Z_i 。

由节点②得

$$\dot{I} = \dot{I}_1 - 0.1S \times \dot{U}_1 = \frac{\dot{U}_1}{20\Omega} - \frac{\dot{U}_1}{10\Omega}$$

又

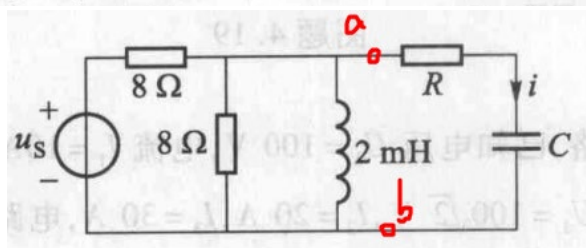
$$\dot{U}_{ab} = (20 - j10)\Omega \dot{I}_1 = (20 - j10) \times \frac{\dot{U}_1}{20}$$

得

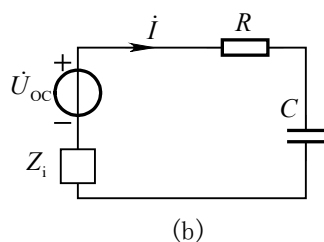
$$Z_i = \frac{\dot{U}_{ab}}{\dot{I}} = \frac{(20 - j10) \times \frac{\dot{U}_1}{20}}{(\frac{1}{20} - \frac{1}{10})\dot{U}_1} = 22.36\angle 153.43^\circ \Omega$$

答案 4.14

4.14 图中 u_s 为正弦电压源, $\omega = 2000 \text{ rad/s}$ 。问电容 C 为何值才能使电流 i 的有效值达到最大?



解: 先对电路 ab 端左侧电路作戴维南等效, 如图(b)所示。



令

$$X_L = \omega L = 2000 \text{ rad/s} \times 2 \times 10^{-3} \text{ H} = 4 \Omega$$

得等效阻抗

$$Z_i = 8 \Omega // 8 \Omega // j4 \Omega = \frac{4 \Omega \times j4 \Omega}{4 \Omega + j4 \Omega} = 2(1 + j) \Omega$$

由

$$i = \frac{U_{oc}}{Z_i + R + \frac{1}{j\omega C}}$$

知, 欲使电流 i 有效值为最大, 电容的量值须使回路阻抗虚部为零, 即:

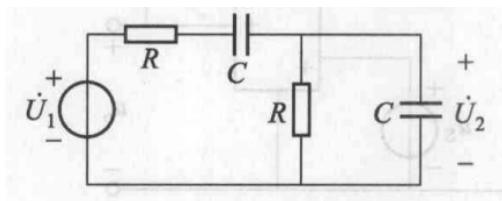
$$\text{Im}[Z_i + R + \frac{1}{j\omega C}] = 2 - \frac{1}{\omega C} = 0$$

解得

$$C = \frac{1}{2\omega} = 250 \mu\text{F}$$

答案 4.18

4.18 图示 RC 分压电路,求频率为何值时 $\dot{U}_2$ 与 $\dot{U}_1$ 同相?



解: 由分压公式得

$$\begin{aligned}\frac{\dot{U}_2}{\dot{U}_1} &= \frac{R // \frac{1}{j\omega C}}{R + \frac{1}{j\omega C} + (R // \frac{1}{j\omega C})} = \frac{\frac{R \times \frac{1}{j\omega C}}{R + \frac{1}{j\omega C}}}{R + \frac{1}{j\omega C} + \frac{R \times \frac{1}{j\omega C}}{R + \frac{1}{j\omega C}}} \\ &= \frac{R}{3R + j(\omega R^2 C - 1/\omega C)}\end{aligned}$$

令虚部

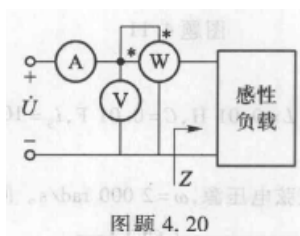
$$\omega R^2 C - \frac{1}{\omega C} = 0, \text{ 得 } \omega = \frac{1}{RC}$$

即 $f = \frac{\omega}{2\pi} = \frac{1}{2\pi RC}$ 时, 且 $\dot{U}_1$ 与 $\dot{U}_2$ 同相位

$$\frac{\dot{U}_2}{\dot{U}_1} = \frac{1}{3}$$

答案 4.20

4.20 图为三表法测量负载等效阻抗的电路。现已知电压表、电流表、功率表读数分别为 36 V、10 A 和 288 W, 各表均为理想仪表, 求感性负载等效阻抗 Z 。再设电路角频率为 $\omega = 314 \text{ rad/s}$, 求负载的等效电阻和等效电感。



图题 4.20

因为 $P = UI \cos \phi$, 所以 $\cos \phi = \frac{P}{UI} = \frac{288}{36 \times 10} = 0.8$

$$|Z| = \frac{U}{I} = 3.6 \Omega$$

$$R = |Z| \cos \phi = 2.88 \Omega$$

$$X_L = |Z| \sin \phi = 1.96 \Omega$$

所以, $Z = (2.88 + j1.96) \Omega$

$$R = 2.88 \Omega$$

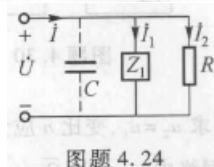
$$L = \frac{1.96}{\omega} = 6.2 \text{ mH}$$

答案 4.24

4.24 图示工频正弦交流电路中, $U=100 \text{ V}$, 感性负载 Z_1 的电流 I_1 为 10 A , 功率因数 $\lambda_1=0.5$, $R=20 \Omega$ 。

(1) 求电源发出的有功功率、电流 I 和总功率因数 λ 。

(2) 当电流 I 限制为 11 A 时, 应并联最小多大电容 C ? 并求此时总功率因数 λ 。



图题 4.24

(1)

$$P = UI_1 \lambda_1 + \frac{U^2}{R} = 1000 \text{ W}$$

假设 $\dot{U} = 100 \angle 0^\circ$

$$\lambda_1 = \cos(\phi_U - \phi_{I_1}) = \cos(0^\circ - \phi_{I_1}) = 0.5$$

所以, $\phi_{I_1} = -60^\circ$

$$\dot{I} = \dot{I}_1 + \dot{I}_2 = 10 \angle (-60^\circ) + \frac{100 \text{ V}}{20 \Omega} \angle 0^\circ = 13.23 \angle 40.89^\circ \text{ A}$$

所以, $I = 13.23 \text{ A}$

$$\lambda = \cos(\phi_U - \phi_I) = \cos(40.89^\circ) = 0.756$$

(2)

负载消耗的总有功功率不变, 总功率因数 λ 为

$$\lambda = \frac{P}{UI} = \frac{1000}{100 * 11} = 0.909$$

当电路仍为感性时, 并联的电容最小, 此时电压超前电流的相位差为:

$$\phi = \arccos(\lambda) = 24.62^\circ$$

电源发出的无功功率

$$Q = P * \tan \phi = 458.26 \text{ var}$$

之前电源的无功功率为

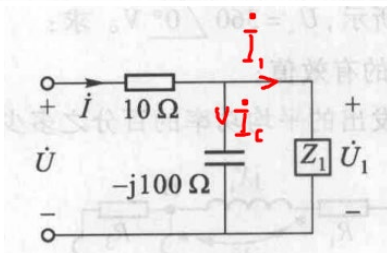
$$Q_1 = UI_1 \sin \phi_1 = 100 * 10 * \sin 60^\circ = 866 \text{ var}$$

无功功率守恒知 $Q = Q_1 + (-\omega C U^2)$

$$C = \frac{Q_1 - Q}{\omega U^2} = \frac{866 - 458.26}{314 * 100^2} = 1.299 * 10^{-4} \text{ F}$$

答案 4.27

4.27 图示电路, $U_1 = 200 \text{ V}$, Z_1 吸收的平均功率 $P_1 = 800 \text{ W}$, 功率因数 $\lambda = 0.8$ (感性)。求电压有效值 U 和电流有效值 I 。



解:

$$\text{设 } \dot{U}_1 = 200 \angle 0^\circ \text{ V}, \varphi_1 = \arccos 0.8 = 36.86^\circ$$

$$I_1 = \frac{P_1}{U_1 \lambda} = 5 \text{ A}, \dot{I}_1 = I_1 \angle -\varphi_1 = 5 \angle -36.86^\circ \text{ A}$$

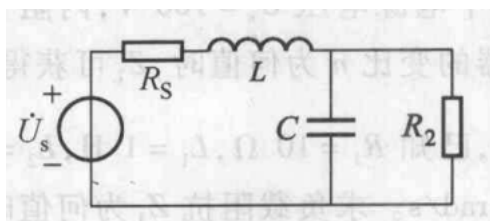
$$\dot{I}_C = \dot{U}_1 / (-j100 \Omega) = j2 \text{ A}, \dot{I} = \dot{I}_C + \dot{I}_1 = (4 - j) \text{ A} = 4.12 \angle -14.04^\circ$$

$$\dot{U} = 10 \dot{I} + \dot{U}_1 = (240 - j10) \text{ V} = 240.2 \angle -2.39^\circ$$

$$I = 4.12 \text{ A}, U = 240.2 \text{ V}$$

答案 4.29

4.29 图示电路中电源频率 $f = 31.8 \text{ kHz}$, $U_s = 1 \text{ V}$, 内阻 $R_s = 125 \Omega$, 负载电阻 $R_2 = 200 \Omega$ 。为使 R_2 获得最大功率, L 和 C 应为多少? 求出此最大功率。



解: L 、 C 及 R_2 的等效阻抗

$$Z_L = j\omega L + \frac{R_2 / (j\omega C)}{R_2 + 1/(j\omega C)}$$

当 L 、 C 改变时, Z_L 的实部及虚部均发生变化, 根据最大功率传输定理知,

当 $Z_L^* = R_s$, R_2 可获得最大功率,

即

$$\begin{cases} \frac{R_2}{1 + (\omega R_2 C)^2} = R_s \\ \omega L - \frac{\omega R_2^2 C}{1 + (\omega R_2 C)^2} = 0 \end{cases}$$

联立解得

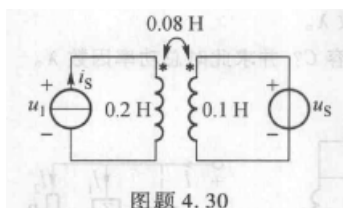
$$\begin{cases} C = \frac{\sqrt{R_2/R_s - 1}}{\omega R_2} = 0.0194 \mu\text{F} \\ L = R_2 R_s C = 0.485 \text{mH} \end{cases}$$

此时

$$P_{\max} = \frac{U_s^2}{4R_s} = \frac{1\text{V}}{4 \times 125\Omega} = 2\text{mW}$$

答案 4.30

4.30 图示电路已知 $i_s = 0.6e^{-10t}$ A, $u_s = 10te^{-20t}$ V。求电压 u_1 的变化规律。



图题 4.30

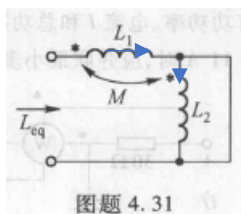
$$u_1 = L_1 \frac{di_s}{dt} + M \frac{di_2}{dt}$$

$$u_s = M \frac{di_s}{dt} + L_2 \frac{di_2}{dt}$$

所以, $u_1 = L_1 \frac{di_s}{dt} + \frac{M}{L_2} \left(u_s - M \frac{di_s}{dt} \right) = (8te^{-20t} - 0.816e^{-10t})\text{V}$

答案 4.31

4.31 求图示电路的等效电感。



图题 4.31

假设端口电压为 u ，端口电流 i ；

L_1 两端电压 $u_1 = u$ ，经过的电流 i ；

L_2 两端电压 $u_2 = 0$ ，经过的电流 i_2 。

$$u = u_1 = L_1 \frac{di}{dt} + M \frac{di_2}{dt}$$

$$u_2 = M \frac{di}{dt} + L_2 \frac{di_2}{dt} = 0$$

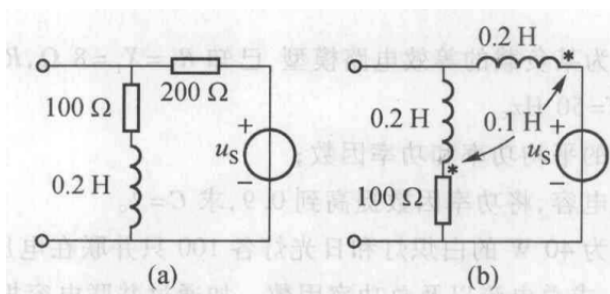
可得

$$u = L_1 \frac{di}{dt} + M \left(-\frac{M}{L_2} \frac{di}{dt} \right) = \left(L_1 - \frac{M^2}{L_2} \right) \frac{di}{dt}$$

$$L_{eq} = L_1 - \frac{M^2}{L_2}$$

答案 4.33

4.33 设图示一端口网络中 $u_s = 200\sqrt{2} \cos \omega t$ V, $\omega = 10^3$ rad/s。求其戴维宁等效电路。



解：(a) 对图(a)电路，感抗

$$X_L = \omega L = 10^3 \text{ rad/s} \times 0.2 \text{ H} = 200 \Omega$$

由分压公式得端口开路电压

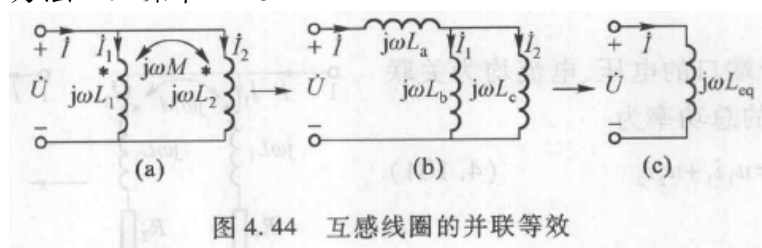
$$\dot{U}_{oc} = \frac{(100 + j200) \Omega}{(100 + j200 + 200) \Omega} \times 200 \angle 0^\circ \text{ V} = 124 \angle 29.7^\circ \text{ V}$$

求等效阻抗，将电压源作用置零

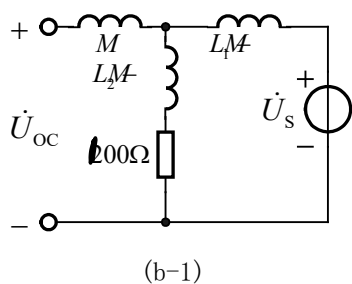
$$Z_i = (100 + j200) \Omega // 200 \Omega = \frac{200 \Omega \times (100 + j200) \Omega}{(200 + 100 + j200) \Omega} = 124 \angle 29.7^\circ \Omega$$

(b)

方法一：课本 P115



对图(b)电路，应用互感消去法，将电路等效成图(b-1)。



图中

$$M = 0.1\text{H}, L - M = 0.1\text{H}。$$

由分压公式得

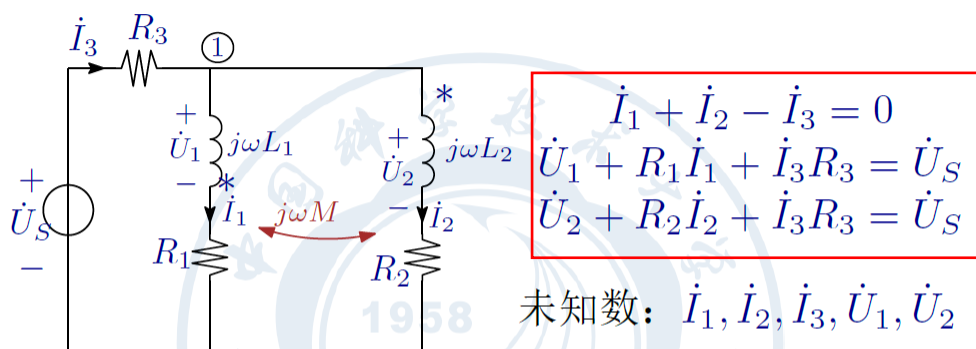
$$\dot{U}_{oc} = \frac{R + j\omega(L_2 - M)}{R + j\omega(L_2 - M) + j\omega(L_1 - M)} \dot{U}_s = (120 - 40j)\text{V} = 126.5 \angle -18.43^\circ \text{V}$$

等效阻抗

$$\begin{aligned} Z_i &= j\omega M + [R + j\omega(L_2 - M)] // j\omega(L_1 - M) \\ &= j\omega M + \frac{[R + j\omega(L_2 - M)] \times j\omega(L_1 - M)}{R + j\omega(L_2 - M) + j\omega(L_1 - M)} = (20 + j160)\Omega = 161.24 \angle 82.87^\circ \Omega \end{aligned}$$

方法二：课件 P35

含互感元件的正弦电流电路



利用相量形式的互感线圈电压电流关系，特别注意同名端的电流流向与 M 的符号关系！

- 注意电流从端口 1 同名端流出，从端口 2 同名端流入，因此 M 选负号。前提是 $\dot{U}_1, \dot{I}_1, \dot{U}_2, \dot{I}_2$ 是关联参考方向。

$$\dot{U}_1 = j\omega L_1 \dot{I}_1 - j\omega M \dot{I}_2, \dot{U}_2 = -j\omega M \dot{I}_1 + j\omega L_2 \dot{I}_2$$

开路电压：

$$\dot{U}_{oc} = \frac{R + j\omega L_1 + j\omega M}{R + j\omega L_1 + j\omega M + j\omega L_2} \dot{U}_S$$

等效阻抗计算：

$$\begin{cases} \dot{I}_3 = \dot{I}_1 + \dot{I}_2 \\ j\omega L_1 \dot{I}_1 + j\omega M \dot{I}_2 + R \dot{I}_1 = U \\ j\omega M \dot{I}_1 + j\omega L_2 \dot{I}_2 = U \end{cases}$$

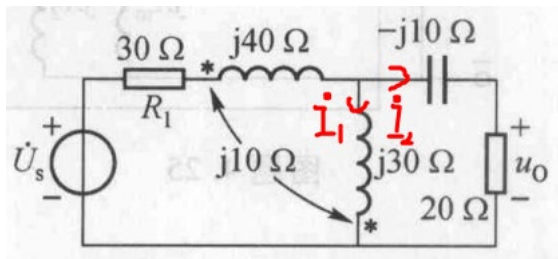
解得 $\dot{I}_1, \dot{I}_2, \dot{I}_3$

$$Z = \frac{U}{\dot{I}_3} = \dots$$

答案 4.35

4.35 电路如图所示, $\dot{U}_s = 360 \angle 0^\circ \text{ V}$ 。求:

- (1) 输出电压 u_o 的有效值;
- (2) 理想电压源发出的平均功率的百分之多少传递到 20Ω 的电阻上?



解:

(1)

$$j30 \cdot \dot{I}_1 - j10 \cdot (\dot{I}_1 + \dot{I}_2) = (-j10 + 20) \cdot \dot{I}_2$$

$$\dot{U}_s = (30 + j40) \cdot (\dot{I}_1 + \dot{I}_2) - j10 \cdot \dot{I}_1 + (-j10 + 20) \cdot \dot{I}_2$$

解得

$$\dot{I}_1 = -j4.5 \text{ A}, \dot{I}_2 = 4.5 \text{ A}$$

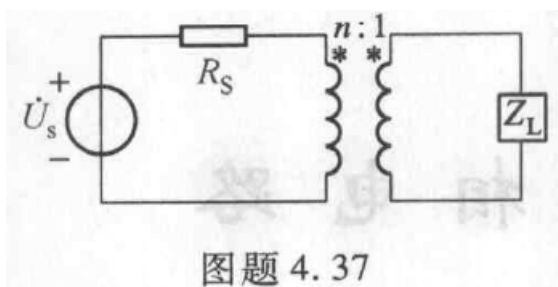
$$\dot{u}_o = 20 \cdot \dot{I}_2 = 90 \text{ V}, \text{ 有效值为 } 90 \text{ V}$$

(2)

$$\eta = \frac{4.5^2 \cdot 20}{360 \cdot 4.5} = \frac{1}{4}$$

答案 4.37

4.37 图示电路中电源电压 $U_s = 100 \text{ V}$, 内阻 $R_s = 5 \Omega$, 负载阻抗 $Z_L = (16 + j12) \Omega$, 问理想变压器的变比 n 为何值时, Z_L 可获得最大功率? 试求此最大功率。



图题 4.37

解: 由理想变压器的阻抗变换关系得

$$Z'_L = n^2 Z_L$$

当变比 n 改变时 Z' 的模改变而阻抗角不变, 此时获得最大功率条件是模匹配, 即

$$R_s = |Z'_L| = |n^2 Z_L|$$

由此求得：

$$n^2 = \frac{R_s}{|Z_L|} = \frac{5\Omega}{\sqrt{16^2 + 12^2}\Omega} = \frac{1}{4}$$
$$n = 0.5$$

设 $\dot{U}_s = 100\angle 0^\circ \text{V}$ ，则理想变压器原端电流：

$$\dot{I}_1 = \frac{\dot{U}_s}{R_s + Z'_L} = \frac{100\angle 0^\circ}{5 + 4 + j3} = \frac{10}{3}\sqrt{10}\angle -18.4^\circ \text{A}$$

副端电流为

$$\dot{I}_2 = -n\dot{I}_1 = -\frac{5}{3}\sqrt{10}\angle -18.4^\circ \text{A}$$

负载吸收的最大平均功率为

$$P_{\max} = I_2^2 \times 16\Omega = \left(\frac{5\sqrt{10}}{3}\right)^2 \times 16 = 444.44 \text{W}$$

重点

1. 相量的写法，注意题目要求，最后是否要转换到时域
2. 互感的方向